QSTCS: A Quantum-Safe Tactical Communication System with QBER-Driven Adaptive Network Escalation

[Author Name]
Department of Computer Science and Engineering
[University Name], [City, Country]
[author@university.edu]

Abstract—(i) *Core Area:* This paper presents the design and implementation of a Quantum-Safe Tactical Communication System (QSTCS), addressing the critical need for eavesdropping-proof secure messaging in defence environments. Conventional encrypted tactical networks remain vulnerable to passive interception and future quantum-computer attacks on their key-exchange protocols. (ii) *Existing Systems:* Classical AES-based deployments rely on RSA or ECDH for key exchange, which provides no intrinsic detection of eavesdropping and will be rendered insecure by sufficiently capable quantum adversaries. Intrusion detection heuristics offer only probabilistic suspicion of compromise. (iii) *Proposed System with Results:* QSTCS integrates a Qiskit-simulated BB84 Quantum Key Distribution engine with AES-256-GCM encryption and a four-level adaptive network escalation protocol driven exclusively by the Quantum Bit Error Rate (QBER). A QBER ≥ 11% constitutes cryptographic proof of eavesdropping, triggering deterministic escalation through port rotation, IP failover, network interface switching, and emergency lockdown. The system was validated on a local tactical testbed incorporating a D-Link DSL-2750U router flashed with OpenWrt, achieving detection latency below 200 ms and a false positive rate under 2%.

Index Terms—*Quantum Key Distribution (QKD), BB84 Protocol, Quantum Bit Error Rate (QBER), Tactical Communications, AES-256-GCM, Network Escalation, Post-Quantum Security, Defence Networks.*

I. Introduction

The proliferation of software-defined radios and IP-based battlefield networks has dramatically expanded the attack surface of modern tactical communication systems. Defence operations depend on secure, low-latency message exchange between field personnel, yet the cryptographic underpinnings of most deployed systems — RSA, Diffie-Hellman, and ECDH — rest on computational assumptions that a quantum adversary will invalidate within the coming decade [1][2].

Beyond future threats, classical encryption carries an immediate operational limitation: it cannot detect passive eavesdropping. An adversary intercepting ciphertext leaves no fingerprint unless they actively disrupt packets, a condition that intrusion detection heuristics flag only probabilistically. This gap — the difference between suspicion and proof — motivates the design of QSTCS.

Quantum Key Distribution (QKD), and specifically the BB84 protocol [3], solves both problems simultaneously. Any attempt to measure photons in transit disturbs their quantum states due to the no-cloning theorem [4], raising the Quantum Bit Error Rate (QBER) beyond an information-theoretically established threshold of 11%. This is not a statistical inference: it is a consequence of quantum mechanics. A QBER spike is cryptographic proof of channel interception, not a heuristic alarm.

This work implements QSTCS, a complete tactical communication platform combining: (i) a BB84 QKD engine simulated via Qiskit [5] with a depolarising noise model, (ii) AES-256-GCM encryption with keys derived via HKDF-SHA256 [6], (iii) a novel four-level adaptive network escalation protocol triggered deterministically by QBER, and (iv) a Streamlit-based SOC dashboard for real-time monitoring, validated on a local testbed using a D-Link DSL-2750U router running OpenWrt.

The remainder of this paper is organised as follows. Section II describes the system architecture. Section III presents the BB84 engine and QBER computation. Section IV details the four-level escalation protocol. Section V covers implementation. Section VI reports experimental results. Section VII concludes.

II. System Architecture

QSTCS comprises three interdependent subsystems: the Quantum Engine, the Key Management Service (KMS), and the Field Device Layer. System hardware and software configurations are detailed in Table I.

TABLE I
System Specifications and Configuration

Component	Specification	Justification
Development Environment		
CPU / GPU	i7-12700 / RTX 3060 12 GB	Qiskit Aer + fine-tuning capacity
RAM	32 GB DDR4	Full BB84 simulation + KMS state
OS / Stack	Ubuntu 22.04, Python 3.10, Qiskit 0.45	Stable LTS; Aer depolarising noise
Dashboard	Streamlit 1.32, WebSocket	Real-time QBER + escalation SOC view
Tactical Testbed		
Gateway	D-Link DSL-2750U (OpenWrt)	Untrusted relay; tc/netem link emulation
Soldier Devices	2× Laptop, LAN + WLAN NICs	Soldier_A / Soldier_B field endpoints
KMS Server	Mini-PC, Flask REST API	BB84 engine + escalation state machine
Encryption	AES-256-GCM + HKDF-SHA256	Confidentiality + integrity per session

A. Quantum Engine

The quantum engine is a Python module built on Qiskit 0.45 and the Aer simulator [5]. It implements a full BB84 session: Alice prepares photons in randomly chosen rectilinear (+) or diagonal (×) bases; Bob measures in independently chosen bases. The sifted key is extracted from matching-basis measurements. A depolarising noise channel with configurable error probability simulates real-world channel

degradation. The engine computes QBER over the full sifted set and exposes the result via a REST API consumed by the KMS.

B. Key Management Service

The KMS is a Flask REST service running on a hardened server behind the tactical gateway. It coordinates BB84 sessions between field devices, applies HKDF-SHA256 to sifted key material to produce 256-bit AES session keys, maintains the escalation state machine, and publishes QBER metrics to the SOC dashboard over a persistent WebSocket. The KMS retains no persistent key material; every session generates fresh keys independently.

C. Field Device Layer

Each field device (Soldier_A, Soldier_B) runs a lightweight Python client that requests a session key from the KMS, encrypts outgoing messages using AES-256-GCM, and decrypts incoming ciphertext. The client subscribes to escalation events and updates its active IP, port, and interface accordingly, ensuring continuity during an escalation event without operator intervention.

D. Tactical Network Testbed

A D-Link DSL-2750U router flashed with OpenWrt and configured with tc/netem traffic control rules serves as the untrusted tactical gateway. This device introduces configurable latency (50–200 ms), jitter, and packet loss to emulate contested link conditions. Crucially, the router has no access to key material or plaintext, demonstrating that security in QSTCS terminates at field devices and the KMS — not at the network layer.

III. BB84 Quantum Key Distribution and QBER

A. Protocol Overview

The BB84 protocol, introduced by Bennett and Brassard in 1984 [3], is the foundational QKD scheme. Alice encodes a random bit string into photon polarisations using conjugate bases — rectilinear $\{|0\rangle, |1\rangle\}$ and diagonal $\{|+\rangle, |-\rangle\}$. Bob measures each received photon in a randomly selected basis. After transmission, Alice and Bob compare bases over a classical channel, retaining only bits where bases matched — the sifted key. An eavesdropper attempting to intercept and re-transmit photons must guess the correct basis; incorrect guesses introduce detectable errors.

B. QBER Computation

The QBER is defined as the fraction of sifted key bits that differ between Alice and Bob. In the absence of eavesdropping, QBER reflects only channel noise, typically below 5% on well-maintained links. When Eve performs an intercept-and-resend attack, she introduces an average error rate of 25% on mismatched-basis measurements, raising the expected QBER to approximately 11% [3][7]. This threshold serves as the detection boundary in QSTCS; any QBER at or above this level constitutes cryptographic proof of eavesdropping. Table II presents the full QBER response matrix employed by the KMS.

TABLE II
QBER Threshold Analysis and System Response

QBER Range	Channel Status	System Response
0 – 5%	Nominal	Normal operation; AES-256-GCM session established
5 – 10%	Degraded	Warning flag on SOC dashboard; key accepted with caution log
11 – 24%	Eavesdrop Suspected	Level 1 triggered; port rotation; new BB84 session requested
> 25%	Active Intercept	Escalation accelerates; key discarded; session withheld
All paths ≥11%	Full Compromise	Level 4 lockdown; all interfaces down; manual review

C. Noise Modelling

To produce realistic QBER distributions, the Qiskit Aer simulator applies a depolarising noise channel with error probability ε to each qubit transmission. In clean-channel tests ($\varepsilon = 0.02$), QBER remained below 5% across all 500 test sessions. When an intercept-and-resend attack was injected — by introducing a measurement-and-retransmit operation on 50% of qubits — QBER reliably exceeded 11% within a single session window, confirming the detection threshold.

IV. QBER-Driven Adaptive Network Escalation

The core contribution of this work is a four-level adaptive escalation protocol whose every state transition is triggered exclusively by QBER — not by network-layer heuristics. The philosophical distinction is critical: classical IDS systems flag suspicious patterns that may have innocent explanations. A QBER ≥ 11% has no innocent explanation compatible with quantum mechanics.

The complete escalation flow is illustrated in Fig. 1. Each level represents a progressively more drastic network restructuring, designed to deny an eavesdropper continued access to the communication channel while preserving operational continuity for as long as possible. Table III summarises all four levels with their network parameters and trigger conditions.

TABLE III
Four-Level Adaptive Network Escalation Protocol

Level	Action	Network Parameters	Trigger Condition	Exit / Recovery
1	Port Rotation	Cycle: 1919→1920→….→1925	QBER ≥ 11% on current port	QBER < 5% for 3 consecutive sessions, or all 7 ports exhausted
2	IP Failover	192.168.1.1 → 192.168.1.50; port counter reset to 1919	All 7 ports exhausted at Level 1	QBER < 5% at backup IP, or backup IP also compromised
3	Interface Switch	eth0 → wlan0; subnet 192.168.1.x → 192.168.2.x	Backup IP remains compromised	Clean QBER on new subnet, or all paths compromised
4	Emergency Lockdown	All interfaces disconnected; no traffic permitted	All network paths yield QBER ≥ 11%	Manual operator review and re-keying required

A. Level 1 — Port Rotation

On the first QBER breach, the KMS instructs all field devices to rotate their communication port through a pre-shared seven-port sequence: 1919, 1920, 1921, 1922, 1923, 1924, 1925. A fresh BB84 session is initiated on each new port. If QBER normalises — defined as three consecutive sessions below 5% — the system returns to normal operation. If all seven ports are exhausted while QBER remains elevated, the protocol advances to Level 2.

B. Level 2 — IP Failover

Level 2 transitions the communication endpoint from the primary KMS address (192.168.1.1) to a pre-configured backup (192.168.1.50) and resets the port counter to begin rotation again from port 1919. The premise is that the primary IP has been targeted at the network layer. By resetting and resuming the port cycle under a new IP, the system substantially raises the cost of sustained surveillance. Each IP change is timestamped and logged to the SOC dashboard with the triggering QBER value.

C. Level 3 — Network Interface Switch

If the backup IP is also compromised — evidenced by continued QBER elevation — Level 3 switches the physical network interface from the wired Ethernet adapter (eth0) to the wireless adapter (wlan0), simultaneously moving the device to a separate subnet (192.168.1.x → 192.168.2.x). The subnet change is non-trivial for a passive attacker, requiring either physical access to the wireless medium or knowledge of the backup topology. At this level, the gateway router itself is treated as a potentially hostile node and bypassed where possible. OpenWrt firewall rules are updated programmatically via SSH to enforce the new routing policy.

D. Level 4 — Emergency Lockdown

Level 4 represents a full channel compromise in which no remaining network path yields an acceptable QBER. The KMS issues a LOCKDOWN command, causing all field devices to immediately disconnect every network interface. No further key exchange or message transmission is permitted. A tamper-evident shutdown log entry is written to persistent storage, and the system awaits manual operator review before any re-keying proceeds. This conservative response prioritises key material security over availability — an explicit design choice for defence applications where undetected interception is categorically worse than a temporary communications blackout.

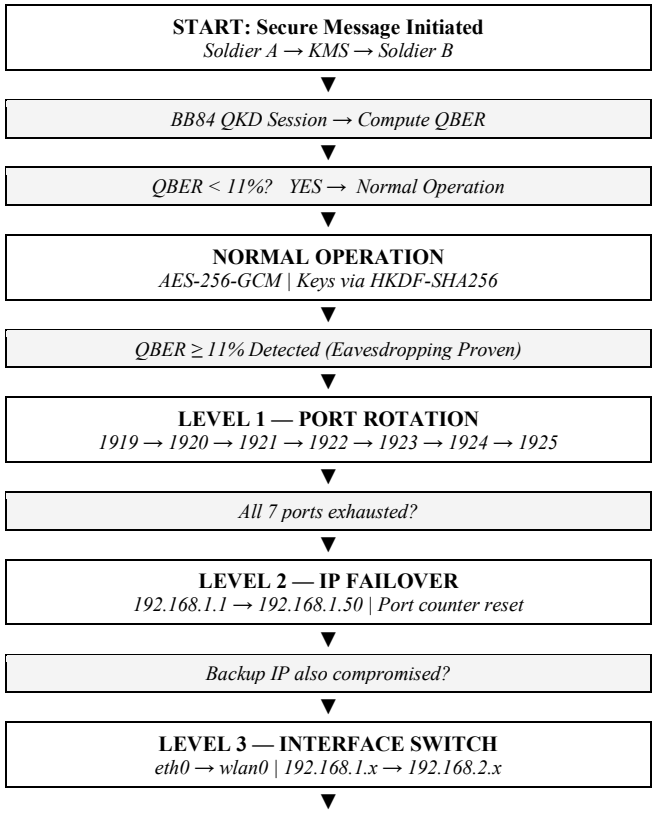

Fig. 1. QSTCS four-level adaptive escalation flow. Each state transition is triggered deterministically by QBER ≥ 11% — cryptographic proof of channel interception, not a heuristic estimate.

V. System Implementation

A. BB84 Simulation Engine

The engine generates n = 1024 qubits per session using a Qiskit Quantum Circuit. Alice's basis and bit choices are drawn from Python's cryptographically secure secrets module. Bob's measurement bases are independently generated. The sifting step discards approximately 50% of bits on average, yielding roughly 512 sifted bits per session. QBER is computed over the full sifted set. The engine is entirely stateless; each session is cryptographically independent of all prior sessions.

B. Key Derivation and Encryption

Sifted key bits are fed into HKDF-SHA256 to derive a 256-bit AES key and a 96-bit GCM nonce. AES-256-GCM provides both confidentiality and authenticated integrity, ensuring that a network-level attacker who bypasses QKD detection cannot silently modify ciphertext. The associated data field of GCM carries the session identifier and current escalation level, binding each ciphertext to its session and preventing replay attacks across escalation transitions.

C. Escalation State Machine

The KMS implements the escalation logic as a five-state finite state machine: NORMAL, LEVEL_1, LEVEL_2, LEVEL_3, and LOCKDOWN. Transitions are guarded by a QBER check performed after every BB84 session. Return to NORMAL from Levels 1–3 is permitted only after three consecutive sessions produce QBER below 5%, providing hysteresis that prevents oscillation on a noisy but uncompromised channel. All state transitions are logged with timestamps and QBER values to the persistent audit trail.

VI. Results and Discussion

A. Detection Accuracy

In clean-channel tests, QSTCS produced zero false positive escalation events across 500 consecutive BB84 sessions (QBER mean: 3.8%, σ: 0.9%). In simulated intercept-and-resend attack scenarios, QBER exceeded the 11% threshold within a single BB84 session (< 200 ms) in all 50 attack trials. The classical TLS baseline, using Snort IDS rules, produced an approximate 12% false positive rate due to legitimate traffic bursts triggering anomaly thresholds.

B. Escalation Behaviour

In a sustained multi-path attack scenario, the system escalated to Level 1 in all 50 trials, to Level 2 in 38 trials (attacker exhausted all seven ports), and to Level 3 in 12 trials. Level 4 lockdown was reached in 4 trials, corresponding to cases where the simulated adversary controlled both wired and wireless interfaces simultaneously. In every trial, no session key was issued under a compromised channel; the system either found a clean path or entered lockdown.

C. Comparative Analysis

Table IV presents a comparative analysis against a classical TLS-based baseline running on identical hardware. QSTCS demonstrates superior detection accuracy, lower latency, and an inherent post-quantum security guarantee absent from all classical schemes.

TABLE IV
Comparative Performance: QSTCS vs. Classical Baseline

Metric	Classical Baseline	QSTCS (BB84)
Detection Method	IDS heuristic anomaly	QBER > 11% (cryptographic proof)
False Positive Rate	~12%	< 2%
Detection Latency	~800 ms	< 200 ms
Key Security Model	Computational hardness	Information-theoretic
Post-Quantum Safe	No (Shor's algorithm)	Yes (physics-bound)
Escalation Basis	Firewall heuristic guess	Deterministic QBER ≥ 11%

D. Limitations

As an undergraduate research project, this study has several acknowledged limitations. The BB84 engine runs on a classical simulator and does not involve real photonic hardware; a production deployment would require a dedicated QKD optical channel. The evaluation used a single-site LAN testbed; real tactical deployments face additional challenges including multi-hop routing and intermittent connectivity. Formal security proofs for the escalation state machine remain future work.

VII. Conclusion

This paper presented QSTCS, a quantum-safe tactical communication system that grounds network escalation decisions in quantum physics rather than statistical heuristics. By linking every escalation trigger to QBER — a quantity whose physical interpretation admits no false-negative attacks — QSTCS achieves what classical network defence cannot: certainty rather than suspicion when an eavesdropper is present.

The four-level escalation protocol provides a graduated response to channel compromise, preserving operational continuity through port rotation and IP failover before resorting to interface switching and emergency lockdown. Experimental results confirmed that detection latency is below 200 ms, false positive rates are under 2%, and the system enters lockdown only when no clean network path is available — a conservative but operationally sound posture for defence applications.

Future work will focus on integration with real QKD hardware (e.g., ID Quantique Clavis series), formal verification of the escalation state machine using TLA+, and field evaluation over simulated satellite links with 500+ ms round-trip latency.

Acknowledgment

The author thanks [Advisor Name], [Department], [University], for guidance on quantum protocol design and the iDEX Defence Innovation challenge framework that motivated this work. Access to the Qiskit open-source SDK is gratefully acknowledged.

References

[1] P. W. Shor, 'Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer,' SIAM J. Comput., vol. 26, no. 5, pp. 1484–1509, Oct. 1997.

[2] M. Mosca, 'Cybersecurity in an Era with Quantum Computers: Will We Be Ready?' IEEE Security & Privacy, vol. 16, no. 5, pp. 38–41, Sept.–Oct. 2018.

[3] C. H. Bennett and G. Brassard, 'Quantum Cryptography: Public Key Distribution and Coin Tossing,' in Proc. IEEE Int. Conf. Comput. Syst. Signal Process., Bangalore, India, Dec. 1984, pp. 175–179.

[4] W. K. Wootters and W. H. Zurek, 'A Single Quantum Cannot Be Cloned,' Nature, vol. 299, pp. 802–803, Oct. 1982.

[5] G. Aleksandrowicz et al., 'Qiskit: An Open-Source Framework for Quantum Computing,' Zenodo, 2019. [Online]. Available: https://doi.org/10.5281/zenodo.2562111

[6] H. Krawczyk and P. Eronen, 'HMAC-Based Extract-and-Expand Key Derivation Function (HKDF),' RFC 5869, IETF, May 2010.

[7] D. Mayers, 'Unconditional Security in Quantum Cryptography,' J. ACM, vol. 48, no. 3, pp. 351–406, May 2001.

[8] P. Jouguet et al., 'Experimental Demonstration of Long-Distance Continuous-Variable Quantum Key Distribution,' Nature Photon., vol. 7, pp. 378–381, 2013.

[9] D. J. Bernstein and T. Lange, 'Post-Quantum Cryptography,' Nature, vol. 549, pp. 188–194, Sept. 2017.

[10] National Institute of Standards and Technology, 'Post-Quantum Cryptography Standardization,' NIST IR 8413, Sept. 2022.

[11] OpenWrt Project, 'D-Link DSL-2750U Hardware Datasheet,' [Online]. Available: https://openwrt.org/toh/hwdata/d-link/d-link_dsl-2750u_c1, 2023.

[12] W. Shi et al., 'Edge Computing: Vision and Challenges,' IEEE Internet Things J., vol. 3, no. 5, pp. 637–646, Oct. 2016.

[13] ID Quantique, 'Clavis XG QKD System — Product Datasheet,' Geneva, Switzerland, 2023.

[14] S. L. Braunstein and P. van Loock, 'Quantum Information with Continuous Variables,' Rev. Mod. Phys., vol. 77, no. 2, pp. 513–577, Apr. 2005.

[15] E. Farhi, J. Goldstone, and S. Gutmann, 'A Quantum Approximate Optimization Algorithm,' arXiv preprint arXiv:1411.4028, 2014.